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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IRB: Experimental model and subject details Subjects The study protocol was reviewed and approved by the Ethics Committee of the Department of Medicine , Helsinki University Hospital , and written informed consent was obtained from all subjects before participation .
Consent: Experimental model and subject details Subjects The study protocol was reviewed and approved by the Ethics Committee of the Department of Medicine , Helsinki University Hospital , and written informed consent was obtained from all subjects before participation .
Inclusion and Exclusion Criteria
Extracted Ion Chromatogram) retention time window , 10 mz tolerance , 75 ppm shared and modified peptides were excluded.
Attrition
Seven men were found to have a previously undiagnosed T2D , had not yet received antihyperglycemic treatment , and are considered here as diet-treated.
Sex as a biological variable
We obtained muscle biopsies from 148 men (aged 20-70 years): 24 men with T2D , 14 men with impaired glucose tolerance (either isolated IGT , or a combination of IFG and IGT) , 25 men with impaired fasting glucose (IFG) , and 85 men with normal glucose tolerance (NGT) (Table 1) .
Subject Demographics
Weight: At the time of his first examination his weight was 76 kg , BMI 25.2 kgm2 , his T2D was managed with metformin only , and his HbA1c was 5.9 .
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.

Replication
Six fully independent replicates were performed for each glucose tolerance category .
Code Information
Availability: Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request .
Data Information
Availability: Data and code availability Raw mass spectrometry data reported in this study cannot be deposited in a public repository because of the EU-GDPR regulations .

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
NH4HCO3 Sigma Aldrich CatA6141	NH4HCO3 Sigma Aldrich		
kit Biognosys Ki-3002-2 Software and algorithms SPSS Statistics IBM version 25 RStudio RStudio Team https:www.rstudio.com RStudio version 1.4.1106 R R Core Team https:www.r-project.org R version 4.0.54 ProteinPilot software AB SCIEX Paragon algorithm (v4.5	https:www.rstudio.com		
	RStudio		
	https:www.r-project.org		
Agilent 3100 OFFGEL fractionator Agilent Technologies RRID:SCR_019388 TripleTOF(r	Agilent OFFGEL fractionator Agilent		Agilent 3100 OFFGEL Fractionator (RRID:SCR_018433)(link)
) 6600 Quadrupole Time-Of-Flight instrument Sciex TripleTOF 6600 Eksigent nanoLC with a Turbo V Source Sciex Turbo V Source Q Exactive ESI-quadrupole-orbitrap mass spectrometer Thermo Fisher Scientific RRID:SCR_020571 EASY-nLC	Q Exactive		Thermo Scientific Q Exactive UHMR Hybrid Quadrupole-Orbitrap Mass Spectrometer (RRID:SCR_020571)(link)
1000 nanoflow LC Thermo Fisher Scientific RRID:SCR_014993 zirconia beads BioSpec Products 1 mm beads IPG strips GE Healthcare 24 cm pH 3-10 C18 microspin columns Nest Group HUM			EASY-nLC 1000 Liquid Chromatograph (RRID:SCR_014993)(link)
The proteins were reduced with Tris(2-carboxyethyl)phosphine (TCEP Sigma Aldrich)	TCEP Sigma Aldrich		
The raw MS data were processed with ProteinPilot software using Paragon algorithm (v4.5 , AB SCIEX Shilov et al. , 2007) .	ProteinPilot		
Blood glucose concentration (participants 1-50) was analyzed by glucose dehydrogenase method (Precision-G Blood Glucose Testing System , Medisense , Abbott , IL , USA) , and plasma glucose concentration (participants 51-148) was analyzed by hexokinase method (Roche Modular , F . Hoffman-La Roche Ltd , Rotkreuz , Switzerland) .	Abbott		
The LC-MSMS analysis of phosphorylated peptides was performed on a Q Exactive ESI-quadrupole-orbitrap mass spectrometer coupled to an EASY-	Xcalibur		

nLC 1000 nanoflow LC (Thermo Fisher Scientific) , using Xcalibur version 3.1.66.10 (Thermo Fisher Scientific) .			
The ten most abundant ions with charge states from 2+ to 7+ were selected for subsequent fragmentation (HCD) , and MSMS spectra were acquired with a resolution of 17,500 , with an AGC target value of 5000 , a maximal injection time of 120 ms and the lowest mass fixed at mz 120 in centroid mode.	MSMS		Suggestion: MSMS (RRID:SCR_003532)(link)
Raw data were processed with MaxQuant version 1.6.0.16 (Cox and Mann , 2008) against the human component of the Uniprot Database (release 01_2020 with 20,303 entries) using the Andromeda search engine (Cox et al. , 2011) .	MaxQuant		
	Uniprot		
Functional annotations Gene Ontology annotations were obtained from DAVID bioinformatics resources (Huang et al. , 2009 Huang et al. , 2008) .	DAVID		
Kinase prediction was made using NetworKIN software (Horn et al. , 2014) .	NetworKIN		
Potential differences in clinical variables between the study groups were analyzed by one-way ANOVA followed by Sidaks post hoc test for multiple comparisons (IBM SPSS Statistics , version 25) .	SPSS		Suggestion: SPSS, RRID:SCR_002865(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Data are given as mean +- SEM , one-way ANOVA followed by Sidaks post hoc test for multiple comparisons .	ANOVA
	Sidaks post hoc test
Students t-test was used to detect statistically significant changes between sample groups , and differences with p-value < 0.05 were considered significant .	Students t-test
The ANOVA p-values were multiple hypothesis corrected with FDR (Benjamini-Hochberg) correction , and the proteins for which q-value < 0.05 were subjected to Dunnetts post hoc test with the NGT group as the reference group .	Benjamini-Hochberg
	Dunnetts post hoc test
When analyzing the effect of glucose tolerance status on phosphoproteome (Figure 6 and Table S3) , 2-sided t-tests with Welch approximation for unequal variances were used and differences with a p-value < 0.05 were considered significant .	t-tests

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